Video 07 | Performers and Sequenced Sounds

Massive PSXL instructional series - video script

Runtime estimate	Narration length	Scenes	Format
about 5:16 minutes	734 words at 150 words per minute	13	Screen capture of the plug-in with voice-over; dark theme, interface size 125 percent

Learning goals

- Draw a Performer's 16 steps and understand Steps, Sync, Division, Rate and Smooth.
- Make an arpeggio, a trance gate and a stepped filter.
- Use Patterns, Randomise and pattern files for exact values.

> **NOTE.** Accuracy rule for the narrator and any generated video: use the on-screen names exactly as written in the action column. After **Load Init (Blank)** the synth is silent until routing is connected. Do not compare Massive PSXL to other products.

Scene table

Scene	On-screen action	Narration (word for word)	Visual notes and call-outs
1	Load a factory preset from Sequence, e.g. Minor Arp. Hold a key and open the Perf 1 tab.	A Performer is a sixteen-step sequencer for modulation. Load the factory preset Minor Arp from the Sequence category, hold one key, and open the Perf 1 tab. You hear an arpeggio from a single note, and you can see the bars. Notice how few controls there are. A Performer gives you a lot of rhythm and melody with only five knobs and sixteen bars.	Show the Sequence category in the browser before opening the tab.
2	Point at the 16 bars; one bar is outlined as it plays. Point at Steps, Rate, Smooth, Sync and Division.	Each bar is one step, and the height is the modulation value from zero to one. The outlined bar is the step playing now. Steps sets the loop length, so with eight steps the last eight bars are dimmed. Sync and Division set how long each step lasts. Smooth glides between steps. Sync follows your project tempo. If you turn it off, the Rate knob sets the speed in steps per second, from one tenth of a hertz up to forty hertz.	Add labels over each control as the cursor touches them.
3	Click Drag across the bars to change the pattern; click Patterns to show the menu.	Click or drag across the bars to draw. The Patterns menu has ready-made shapes: ramps, triangle, alternate high and low, random, flat, all zero, and gate on and off pairs. Randomise fills the bars with random values. Smooth is useful for filter movement, because gliding between steps removes the stair-step sound. For pitch it turns the arpeggio into a wobbling slide.	Pause on the Patterns menu for 3 seconds.
4	Load Init (Blank), quick setup Osc 1 > Filter 1 > Out. Osc 1 Pos 0.66, Filter 1 Cutoff 2600, Reso 0.25. Amp Env Attack 0.001, Decay 0.2, Sustain 0.8, Release 0.1.	Now build our own. Load Init Blank, wire Osc 1 into Filter 1 into Out, saw position, cutoff twenty-six hundred, a little resonance, and a short amp envelope with sustain at point eight so the note keeps playing.	Brief time-lapse of the setup.

Scene	On-screen action	Narration (word for word)	Visual notes and call-outs
5	Perf 1 tab: Steps 8, Sync on, Division 1/16, Smooth 0. Drag the Perf 1 chip onto Osc 1 Pitch; right-click > Edit amount +0.25.	Open Perf 1. Eight steps, sync on, sixteenth notes. Drag the Perf 1 chip onto the Osc 1 Pitch knob and set the amount to plus point two five. With that amount a full-height bar is twenty-four semitones, so bar height times twenty-four is the number of semitones. The amount is a fraction of the knob's travel. The Pitch knob covers ninety-six semitones, so plus point two five is twenty-four semitones.	On-screen math: 'height x 24 = semitones'.
6	Draw bars: 0, 0.125, 0.29, 0.5, 0.625, 0.5, 0.29, 0.125. Hold a key.	Draw the bars: nothing, an eighth, just under a third, half, five eighths, half, a third again, and an eighth. That is zero, three, seven, twelve, fifteen, twelve, seven and three semitones. Hold a key and you hear a minor arpeggio climbing and falling. If the highest bar is not quite where you want it, just change the amount, or redraw the bar. Because each note has its own Performer, chords arpeggiate together if you play them at the same moment.	Overlay the semitone numbers above each bar.
7	Show UI scaling: Menu > Settings > Interface size > 175%. Demonstrate drawing more accurately.	Performer pitch steps are continuous, not snapped, so drawing by hand can leave you a few cents out. Make the interface bigger for more precision. For exact values, use Patterns, Save pattern, open the small text file in an editor, type your numbers separated by commas, and use Load pattern. Saving a pattern is also a nice way to build a library of rhythmic gates and melodic lines that you can drop into any patch.	Show the .psxlperf file open in a text editor with the 16 comma-separated values.
8	Add Mod 2 envelope (Attack 0.001, Decay 0.15, Sustain 0.2) to Cutoff +0.25; add Delay Sync on, Division 1/8, Mix 0.25.	Add some sparkle. A short filter envelope on Mod 2 gives each note a pluck at the start, and a tempo-synced delay fills the gaps. Notice that each new note restarts the Performer, so play on the beat. The Performer restarts at note-on, so for tight rhythms play the notes on the grid. The delay in time with the tempo fills the gaps and makes simple patterns sound big.	Overlay a beat grid to show the restart on note-on.
9	New patch: Init (Blank), Osc 1 > Filter 1 > Out, Osc 1 Unison 6 Detune 24, Amp Level 0, Amp Env Sustain 1.0. Perf 1: Steps 16, Division 1/16, Smooth 0.15; Patterns > Gate: on/off pairs.	Now a trance gate. Set the Amp Level knob to zero, and the amp envelope sustain to one. In Perf 1 pick sixteen steps and the Patterns entry gate on and off pairs. Why set Amp Level to zero? Because the Performer adds to the knob position. If the knob were at its default, the gate would never fully close.	Show the Amp Level knob at 0 and the gate pattern bars.
10	Drag Perf 1 chip onto Amp Level; Edit amount +0.75. Edit bars to taste: 1 0 1 1 0 1 0 1 1 0 1 0 1 1 0 0. Hold a chord.	Drag the Performer chip onto Amp Level with plus point seven five. Edit the bars to a rhythm you like. The Performer now opens and closes the level in sixteenth notes. A little Smooth, around point one five, removes clicks.	Show the bars with the rhythm; animate the amp meter pulsing.
11	Third idea: assign Perf 2 to Filter 1 Cutoff; Randomise.	Performers can drive anything. Try Perf 2 on filter cutoff, press Randomise, and you have a stepped filter melody. Other targets: pan, insert mix, or oscillator position. You can use all three Performers at once, at different step counts, to create evolving polyrhythms. A five-step and an eight-step Performer drift in and out of phase.	Quick montage of the target knobs ringing.
12	Save the patterns folder by Patterns > Save pattern... and reveal Presets/Patterns.	Patterns can be saved and loaded as files in your Presets Patterns folder, so you can build a library. The bars also save inside presets. That is Performers.	Show the Patterns folder in Finder.
13	End card.	Next up: drum sounds, built from noise, sines and envelopes.	End card: 'Next: 08 Drum sounds'.

Narration only (plain script for text-to-speech or NotebookLM)

[Scene 1] A Performer is a sixteen-step sequencer for modulation. Load the factory preset Minor Arp from the Sequence category, hold one key, and open the Perf 1 tab. You hear an arpeggio from a single note, and you can see the bars. Notice how few controls there are. A Performer gives you a lot of rhythm and melody with only five knobs and sixteen bars.

[Scene 2] Each bar is one step, and the height is the modulation value from zero to one. The outlined bar is the step playing now. Steps sets the loop length, so with eight steps the last eight bars are dimmed. Sync and Division set how long each step lasts. Smooth glides between steps. Sync follows your project tempo. If you turn it off, the Rate knob sets the speed in steps per second, from one tenth of a hertz up to forty hertz.

[Scene 3] Click or drag across the bars to draw. The Patterns menu has ready-made shapes: ramps, triangle, alternate high and low, random, flat, all zero, and gate on and off pairs. Randomise fills the bars with random values. Smooth is useful for filter movement, because gliding between steps removes the stair-step sound. For pitch it turns the arpeggio into a wobbling slide.

[Scene 4] Now build our own. Load Init Blank, wire Osc 1 into Filter 1 into Out, saw position, cutoff twenty-six hundred, a little resonance, and a short amp envelope with sustain at point eight so the note keeps playing.

[Scene 5] Open Perf 1. Eight steps, sync on, sixteenth notes. Drag the Perf 1 chip onto the Osc 1 Pitch knob and set the amount to plus point two five. With that amount a full-height bar is twenty-four semitones, so bar height times twenty-four is the number of semitones. The amount is a fraction of the knob's travel. The Pitch knob covers ninety-six semitones, so plus point two five is twenty-four semitones.

[Scene 6] Draw the bars: nothing, an eighth, just under a third, half, five eighths, half, a third again, and an eighth. That is zero, three, seven, twelve, fifteen, twelve, seven and three semitones. Hold a key and you hear a minor arpeggio climbing and falling. If the highest bar is not quite where you want it, just change the amount, or redraw the bar. Because each note has its own Performer, chords arpeggiate together if you play them at the same moment.

[Scene 7] Performer pitch steps are continuous, not snapped, so drawing by hand can leave you a few cents out. Make the interface bigger for more precision. For exact values, use Patterns, Save pattern, open the small text file in an editor, type your numbers separated by commas, and use Load pattern. Saving a pattern is also a nice way to build a library of rhythmic gates and melodic lines that you can drop into any patch.

[Scene 8] Add some sparkle. A short filter envelope on Mod 2 gives each note a pluck at the start, and a tempo-synced delay fills the gaps. Notice that each new note restarts the Performer, so play on the beat. The Performer restarts at note-on, so for tight rhythms play the notes on the grid. The delay in time with the tempo fills the gaps and makes simple patterns sound big.

[Scene 9] Now a trance gate. Set the Amp Level knob to zero, and the amp envelope sustain to one. In Perf 1 pick sixteen steps and the Patterns entry gate on and off pairs. Why set Amp Level to zero? Because the Performer adds to the knob position. If the knob were at its default, the gate would never fully close.

[Scene 10] Drag the Performer chip onto Amp Level with plus point seven five. Edit the bars to a rhythm you like. The Performer now opens and closes the level in sixteenth notes. A little Smooth, around point one five, removes clicks.

[Scene 11] Performers can drive anything. Try Perf 2 on filter cutoff, press Randomise, and you have a stepped filter melody. Other targets: pan, insert mix, or oscillator position. You can use all three Performers at once, at different step counts, to create evolving polyrhythms. A five-step and an eight-step Performer drift in and out of phase.

[Scene 12] Patterns can be saved and loaded as files in your Presets Patterns folder, so you can build a library. The bars also save inside presets. That is Performers.

[Scene 13] Next up: drum sounds, built from noise, sines and envelopes.

> **TIP.** Pause for about one second between scenes and let on-screen actions finish before the next sentence starts.